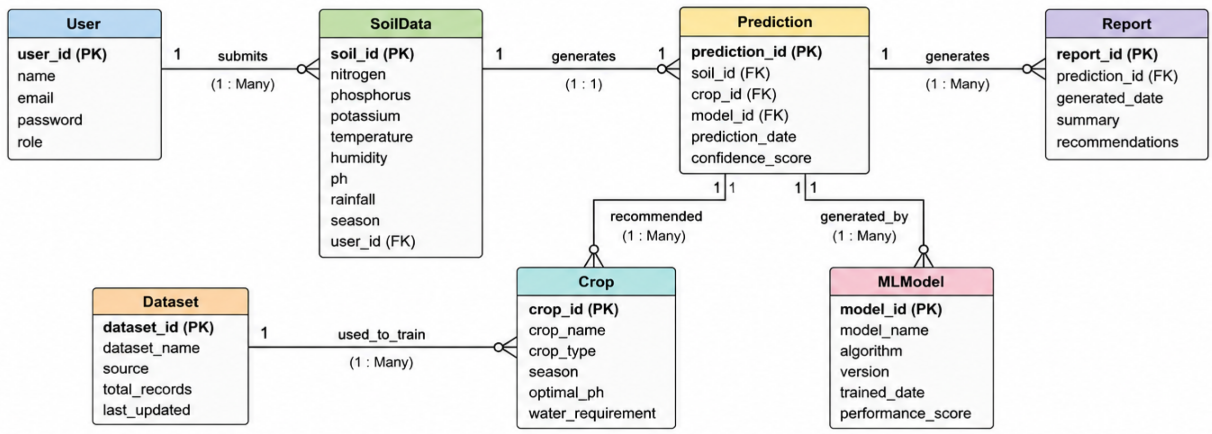
OptiCrop Smart Agricultural Production Optimization System – ER Diagram



DESCRIPTION

The ER diagram represents the core entities involved in the Loan Approval Prediction System and illustrates how they interact with each other. It provides a clear structure for organizing applicant information, loan requests, prediction results, and machine learning model outputs within the database.

ENTITIES INVOLVED

1. User
2. SoilData
3. Crop
4. Dataset
5. MLModel
6. Prediction
7. Report

PRIMARY KEYS

- **User:** user_id
- **SoilData:** soil_id
- **Crop:** crop_id
- **Dataset:** dataset_id
- **MLModel:** model_id
- **Prediction:** prediction_id
- **Report:** report_id

RELATIONSHIPS

- **User to SoilData:** 1 to Many
- **SoilData to Prediction:** 1 to 1
- **Crop to Prediction:** 1 to Many
- **MLModel to Prediction:** 1 to Many
- **Dataset to MLModel:** 1 to Many
- **Prediction to Report:** 1 to Many

FOREIGN KEYS

Foreign Key	References	Purpose
SoilData.user_id	User.user_id	Links soil data to the user
Prediction.soil_id	SoilData.soil_id	Links prediction to soil data
Prediction.crop_id	Crop.crop_id	Links prediction to crop
Prediction.model_id	MLModel.model_id	Links prediction to model
Report.prediction_id	Prediction.prediction_id	Links report to prediction
MLModel.dataset_id	Dataset.dataset_id	Links model to dataset

CARDINALITY

A single user can submit multiple soil data records for agricultural analysis. Each soil data entry is processed by the machine learning model to generate crop prediction results. A crop can appear in multiple predictions, while a single machine learning model can generate predictions for many users and soil conditions. Each prediction can further generate multiple agricultural reports containing summaries and farming recommendations.

NORMALIZATION AND STRUCTURE

The ER diagram is normalized to separate user information, soil parameters, crop details, datasets, machine learning models, prediction records, and reports into different entities. This structure reduces redundancy, improves scalability, and ensures efficient management of agricultural and prediction-related data.

USE CASE COVERAGE

This ER model supports the core functionalities of the OptiCrop Smart Agricultural Production Optimization System, including:

- Collecting and managing soil and environmental data from users.
- Predicting suitable crops using machine learning algorithms.
- Managing datasets and trained machine learning models.
- Generating intelligent crop recommendations and prediction reports.
- Supporting sustainable farming practices and data-driven agricultural decision-making.